

CryptoTrack

Project Overview & Core Functionality

1. Project Overview

CryptoTrack is a robust, full-stack web application designed to empower users with real-time cryptocurrency market insights and portfolio management tools. Built on the MERN stack (MongoDB, Express, React, Node.js), it combines powerful data processing with a modern, glassmorphic user interface to deliver a premium user experience.

2. Core Functionalities

A. Real-Time Market Dashboard

- Live Updates: Real-time price and market data updates using WebSockets (Socket.io).
- Comprehensive Data: View current price, 24h change %, market cap, and volume for top coins.
- Search & Filter: Instantly search for specific cryptocurrencies.

B. Portfolio Management

- Asset Tracking: Users can add their holdings to calculate total portfolio value.
- Profit/Loss Analysis: Automatic calculation of gains and losses based on current market prices.
- Visual Analytics: Interactive charts (e.g., Pie Charts) to visualize portfolio distribution.

C. Watchlist System

- Favorites: Users can star/favorite coins to add them to a personalized watchlist.
- Quick Access: Dedicated view for monitoring interested assets separate from the general market.

D. User Authentication & Security

- Secure Access: JWT (JSON Web Token) based authentication for stateless security.
- Google OAuth: Integrated "Sign in with Google" for seamless onboarding.
- Data Protection: Password hashing with Bcrypt and protected API endpoints.

3. Technical Architecture

The application is built using the following technologies:

- Frontend: React.js (Vite), Tailwind CSS, Framer Motion, Recharts
- Backend: Node.js, Express.js
- Database: MongoDB, Mongoose ODM
- Real-time Communication: Socket.io
- Authentication: Passport.js (Google Strategy), JWT, Bcrypt